



Synthetic Telepathy

Synthetic telepathy is the process of communication using one's thoughts alone. The first step in this direction is the Audeo voice synthesizer developed by Michael Callahan which enables voiceless communication. The device places sensors on the user's neck to capture electrical signals being sent from the brain to the vocal chords, which are then amplified and decoded by the onboard software and played as speech through speakers. However, the device currently requires days of practice, and the resultant speech is also quite slow. In addition, Pentagon's DARPA has initiated the Silent Talk program aiming to enable communication without the use of speech by reading neural signals. The US Army is funding similar research projects in some US universities whose aim is to come up with a computer interface capable of sending messages by use of thought alone. The idea is to develop a helmet which includes an electroencephalograph (EEG) device to read the brain's signals which would then be interpreted by an onboard computer and translated into words. In a recent experiment, a robotic limb was operated by a monkey using one such device. However, if you are getting excited about making creative uses of telepathy, take a back seat, since it will take at least 10 years to come to fruition.

Future of Communication

How we may communicate in the future can be predicted to some extent by looking at some of the technologies being researched right now with a projected implementation 15 years down the line.

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A new year and a new hangover. As Sanket wakes up, he surveys the damage from last night and spots the first casualty of the night before – his beloved 3D phone. It was an old model, but it was in his comfort zone. It had all the basic features – making 3D video calls and playing cheap holographic interactive games –

nothing fancy. Plus, it had all the cool augmented reality apps, handpicked and customised over the last three years. Sanket especially liked the one which could identify faces of the people in front of him and display their Facebook profiles instantly. It really came in handy given his propensity to forget names. Well, at least his afternoons were exciting, given the entertaining experience cellphone shopping usually provides...

While he browsed the shelves of the electronics stores, those annoying audio spotlight advertisements kept coming his way. Considering the kind of nuisance they were, there



Mobile phone implants

In 2002, Royal College of Art designers came up with a tooth implant that consisted of a receiver and transducer. The receiver picks up mobile signals and a transducer converts them to vibrations that travel through jaw bones and is audible only to the user. Since then, there have been several innovations. Similarly, a six-axis piezoelectric accelerometer detects jaw and head motion is used to dial, while you type using hand and jaw gestures. Batteries use a Seebeck-effect strip to convert body heat into electricity. ROM, memory and other components are also implanted similarly and pieces communicate with each other using flexible circuitry and conductive tattoo ink!

Integrating augmented reality with mobiles

Augmented Reality enables you to determine the user's location and displays computer-generated content according to a user's surroundings. It uses GPS and runs image recognition algorithms over the mobile camera images to determine the user's location and then retrieves geographical coordinates and nearby landmarks from an external database. This information can range from prices and comments on local businesses.



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